

Flutter Documentation: Save File From App to Device

Overview

This documentation explains how to save files from a Flutter application directly to the user's device storage. This is commonly used for:

- Saving PDF files
- Downloading images
- Exporting reports
- Storing generated documents

This guide uses: - `path_provider` - `dart:io`

1. Add Required Packages

Add the following dependencies in **pubspec.yaml**.

```
dependencies:  
  flutter:  
    sdk: flutter  
  
  path_provider: ^2.1.2
```

Run:

```
flutter pub get
```

2. Import Required Libraries

```
import 'dart:io';  
import 'package:path_provider/path_provider.dart';
```

3. Get Device Directory

You must first get a directory where the file will be stored.

Example:

```
Future<Directory> getLocalDirectory() async {  
  final directory = await getApplicationDocumentsDirectory();  
  return directory;  
}
```

Common Directories

Method	Description
<code>getApplicationDocumentsDirectory()</code>	Private app storage
<code>getTemporaryDirectory()</code>	Temporary cache storage
<code>getExternalStorageDirectory()</code>	Android external storage

4. Save File To Device

Example function to save a text file.

```
Future<File> saveFile(String fileName, String content) async {  
  final directory = await getApplicationDocumentsDirectory();  
  final file = File('${directory.path}/${fileName}');  
  
  return file.writeAsString(content);  
}
```

Usage:

```
await saveFile('example.txt', 'Hello Flutter File Save');
```

5. Save Binary Files (PDF / Image)

If you have **bytes** instead of text.

```
Future<File> saveBinaryFile(String fileName, List<int> bytes) async {
  final directory = await getApplicationDocumentsDirectory();
  final file = File('${directory.path}/${fileName}');

  await file.writeAsBytes(bytes);
  return file;
}
```

Example:

```
await saveBinaryFile('report.pdf', pdfBytes);
```

6. Show Saved File Path

You can print the saved file location.

```
final file = await saveFile('example.txt', 'Hello');

print(file.path);
```

Example Output:

```
/data/user/0/com.example.app/documents/example.txt
```

7. Example Button to Save File

```
ElevatedButton(
  onPressed: () async {
    await saveFile('test.txt', 'Flutter Save File Example');
  },
  child: Text('Save File'),
)
```

Full Source Code

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:path_provider/path_provider.dart';

class SaveFileExample extends StatelessWidget {

  Future<File> saveFile(String fileName, String content) async {
    final directory = await getApplicationDocumentsDirectory();

    final file = File('${directory.path}/${fileName}');

    return file.writeAsString(content);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Save File Example"),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () async {
            final file = await saveFile(
              'example.txt',
              'Hello Flutter File Save Example',
            );

            debugPrint("Saved at: ${file.path}");
          },
          child: Text("Save File"),
        ),
      ),
    );
  }
}
```

Notes

- Files saved using `getApplicationDocumentsDirectory()` are private to the app.
- To save files visible in the device **Downloads folder**, additional configuration is needed.

- Android 11+ requires scoped storage handling.

Recommended Packages (Advanced)

Package	Purpose
permission_handler	Request storage permission
file_saver	Save files directly to device
open_filex	Open saved files

Summary

Steps to save files in Flutter:

1. Add `path_provider`
2. Get device directory
3. Create a `File`
4. Write data using `writeAsString()` or `writeAsBytes()`
5. Access file using saved path

End of Documentation